

★ PRIMER · UNIT 15

Before the Lecture: Attention and Neighbours

Unit 15 · The Frontier

Watch this first if weighted averages and graphs are hazy

Twenty minutes of groundwork — attention as a weighted average you already understand, and a graph as nothing more than a list of who is connected to whom.

How to Use This Primer

Two audiences, one deck — read the column that describes you

If this is new to you

- Watch before the lecture.
- Attention and graphs are here.
- Do slide 8's arithmetic.

If you have met it before

- Skim it — this is revision.
- Slide 7 is what finance changes.
- Slide 16 says if you can skip.

Nothing here is examined on its own — it is the floor the lecture stands on

Where Unit 15 Sits

The course builds one pipeline; here is this unit's stone

Unit 14

done

Cause, explanation and
fairness.

Unit 15

you are here

Transformers, graphs and
agents.

Unit 16

next

Bring the whole pipeline
together.

Nothing here is new mathematics. It is unit 8's weighted sum, pointed at new shapes.

What Attention Actually Is

Attention is a weighted average where the model decides the weights, one prediction at a time.

An ordinary average

- Every past day counts equally.
- Or a fixed decay.
- The weights are yours.

Attention

- Weights depend on the input.
- Yesterday may dominate today.
- The model chooses.

That is the whole idea. Transformers are attention stacked, at scale, with plumbing.

The Words the Lecture Assumes

Five terms, and both halves of the unit open up

Word	What it means	Here
Attention weight	how much one input counts	which past day matters
Node	one entity in a graph	one company
Edge	a link between two nodes	a supplier relationship
Neighbour	a node one edge away	your direct suppliers
Message passing	neighbours share information	a shock spreading

A graph is a list of nodes and a list of edges. Nothing more mysterious than that.

Three Inputs a TFT Juggles

Most forecasters accept one kind; this is why the architecture exists

Input type	Example	Known ahead?
Past observed	yesterday's return	only afterwards
Known future	earnings date, holiday	yes, in advance
Static	sector, country	never changes

Knowing an earnings date is next Tuesday is information. Classical forecasters throw it away.

Why Stocks Do Not Live Alone

A model that sees each company in isolation cannot see that its only supplier just failed.

Row by row

- Each firm is one row.
- Rows are independent.
- Contagion is invisible.

As a graph

- Firms are nodes.
- Supply links are edges.
- Shocks travel along them.

Finance is full of graphs already — supply chains, ownership, interbank lending, sectors.

One Attention Forecast by Hand

Three past days, three relevance scores — compute the forecast

Day	Value	Score	Weight	Contribution
t-3	10.0	1.0	0.12	1.22
t-2	12.0	2.0	0.33	3.98
t-1	14.0	2.5	0.55	7.65
forecast	—	—	1.00	12.85

Weights are the softmax of the scores, so they sum to one. The forecast is 12.85.

A Shock Spreading, by Hand

One node updates from its neighbours — that is a GNN layer

The setup

- Firm A's value: 0.
- Two neighbours: -6 and -4 .
- A supplier has failed.

One update

- Average the neighbours: -5 .
- Blend with A's own value.
- A now reads -2.5 .

Stack two layers and the shock reaches your suppliers' suppliers

What an AI Agent Adds

A model that can call tools stops being only a predictor

A plain model

- Input goes in, answer out.
- One step, no tools.
- It cannot look anything up.

An agent

- Plans a sequence of steps.
- Calls tools and reads results.
- Revises, then answers.

RAG from unit 12 was the first tool call. An agent chooses which tools, and when.

How a GNN Layer Works

Three moves, and stacking them is the whole architecture

1 · Gather

from neighbours

Collect the current values of every node one edge away.

2 · Aggregate

into one number

Average them, or sum them, or attend over them.

3 · Update

this node

Blend that with the node's own value, through a small network.

Two layers reach two hops. Too many and every node ends up looking identical.

When the Frontier Is Worth It

Be honest — most of the time unit 7's boosted trees still win

Use it when	Because	Otherwise
You have known-future inputs	TFT uses them; trees cannot	use XGBoost
Relationships matter	a graph encodes them	use XGBoost
The task needs many steps	an agent can plan	use one model

The frontier is not an upgrade. It is a different shape of problem, and most are not that shape.

Notation You Will Meet

Four symbols carry the whole unit

Symbol	Read it as	It stands for
α	alpha	an attention weight, summing to one
h	h	a node's current value, or hidden state
$N(i)$	neighbours of i	the nodes one edge from i
softmax	soft max	turns scores into weights, as in unit 12

Softmax appears for the third time. It is how scores become weights, everywhere.

The Python You Will See

You do not need to write it yet — you need to recognise it

Line	What it does
<code>softmax(scores)</code>	turn relevance scores into attention weights
<code>(weights * values).sum()</code>	the weighted average you just did by hand
<code>edge_index</code>	the list of connected node pairs — the graph
<code>GCNConv(in_dim, out_dim)</code>	one message-passing layer
<code>agent.run(task)</code>	plan, call tools, read results, answer

The first two lines are attention in full. The rest of a transformer is scale and plumbing.

Three Things Beginners Get Wrong

Each one costs a week if you carry it into unit 16

1

"Transformers are the state of the art, so use them"

On tabular financial data with a few thousand rows, boosted trees usually still win.

2

"Attention shows what the model thinks is important"

Attention weights are not explanations. Unit 14 already gave you the tool that is.

3

"More GNN layers means more context"

Past two or three, every node converges to the same value. It is called over-smoothing.

If one of these surprised you, that is exactly what the primer was for

Self-Check Before the Lecture

Answer these three out loud — if you can, you are ready

1	Scores 1.0, 2.0, 2.5 on 10, 12, 14. What is the forecast?	12.85
2	Neighbours at -6 and -4. What is the aggregate?	-5
3	What does an agent have that a plain model does not?	tools

Cover the right column, answer, then check. All three come from this deck.

What the Lecture Does With This

You bring the vocabulary; the lecture brings the argument

1

It builds attention from the average

Classical forecasting against a TFT, and the three input types it can juggle.

2

It puts finance on a graph

Nodes, edges and message passing, with a shock spreading through suppliers by hand.

3

It ends with agents

An LLM that plans and calls tools — and an honest account of when none of this is warranted.

Next: Unit 15 lecture — The Frontier